2023 SSMO Accuracy Round

SMO Team

AR 1: Mr. Sammy proposes a Hamburger Proclamation, which has 500 lines, divided into paragraphs of 5 lines each. It takes him 23 seconds to read each line. Additionally, he adds a 0.5 second pause between two lines in a paragraph, and a 2 second pause between paragraphs. If it takes him S minutes to read the whole Hamburger Proclamation, find $\lfloor 10S \rfloor$.

AR 2: Suppose that the average of all n-digit palindromes is denoted by P_n and the average of all n-digit numbers is denoted by N_n. Find the value of $\left\lfloor \sum_{n=1}^{100} (P_n - N_n) \right\rfloor$.

AR 3: Suppose that a, b, c are real numbers such that

$$\begin{aligned} a + b - c &= 4, \\ a^2 + b^2 + c^2 &= 14, \text{ and} \\ a^3 + b^3 - c^3 &= 34. \end{aligned}$$

Find the sum of all possible values of $a + b + c$.

AR 4: In square $ABCD$, point E is selected on diagonal AC. Let F be the second intersection of the circumcircles of triangles ABE and CDE. Given that $AB = 10$ and $EF = 6$, find the maximum possible area of triangle BEC.

AR 5: Define the "relationship" between two positive integers a and b to be $\frac{\tau(ab)}{\tau(a)\tau(b)}$ where $\tau(x)$ is the number of divisors of x. Find the sum of all integers $1 \leq n \leq 100$ which have a relationship of $\frac{3}{4}$ with 360.

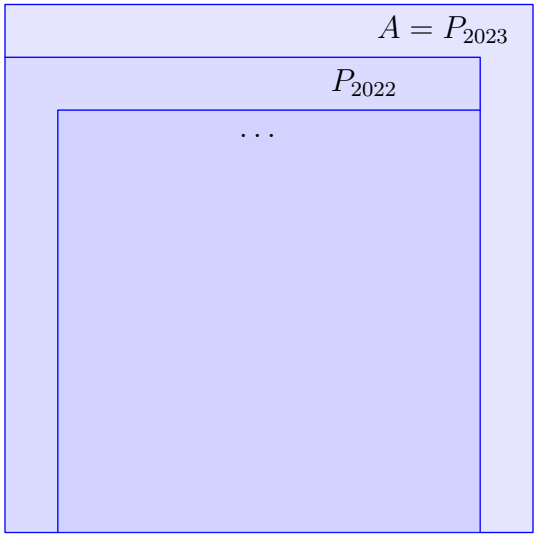
AR 6: Let the roots of $P(x) = x^3 - 2023x^2 + 2023^{2023}$ be α, β, γ. Find

$$\frac{\alpha^2 + \beta^2}{\alpha + \beta} + \frac{\beta^2 + \gamma^2}{\beta + \gamma} + \frac{\gamma^2 + \alpha^2}{\gamma + \alpha}.$$

AR 7: Concentric circles ω and ω_1 are drawn, with radii 3 and 5, respectively. Chords AB and CD of ω_1 are both tangent to ω, and lines AB and CD intersect at P. If $PA = PC = 3$, then the sum of all possible distinct values of $[PAD]$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

AR 8: There is a quadrilateral $ABCD$ inscribed in a circle ω with center O. In quadrilateral $ABCD$, diagonal AC is a diameter of the circle, $\angle BAC = 30°$, and $\angle DAC = 15°$. Let E be the foot of the perpendicular from O to side BA. Let F be the foot of the perpendicular from E to BO. Given that $EF = 3$, the product of the lengths of the diagonals of $ABCD$ can be expressed as $a\sqrt{b}$, where a and b are positive integers and b is not divisible by the square of any prime. Find $a + b$.

AR 9: Consider a 2023×2023 grid called $A = P_{2023}$. We take one of the four smaller 2022×2022 grids located in P_{2023} as P_{2022}. We repeat the process of taking smaller grids until we eventually end at the unit square P_1.



Of the 4^{2022} distinct tuples of shrinking grids $(P_{2023}, P_{2022}, \ldots, P_1)$, let T be the number of these tuples such that their last element is the center square of the original grid A. Find the largest integer a such that $2^a \mid T$.

AR 10: Let $\triangle ABC$ be a triangle such that $AB = 13$, $BC = 14$, $CA = 15$. Let the incircle of $\triangle ABC$ touch BC at D, AC at E, and AB at F. Let ℓ_A be the line through the midpoints of AE and AF. Define ℓ_B and ℓ_C similarly. The area of the star created by the union of $\triangle ABC$ and the triangle bounded by ℓ_A, ℓ_B, and ℓ_C can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.